

Dr. Rachik Soualah
Department of Physics, College of Engineering and Physical Sciences
Khalifa University of Science and Technology
Abu Dhabi, UAE
Phone: + 971 (0) 2 3123196
Email: rachik.soualah@ku.ac.ae
Website: cern.ch/rsoualah

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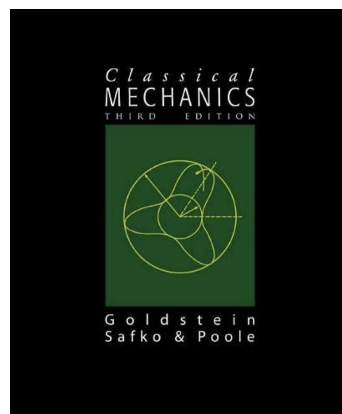
PHYS 725: Classical Mechanics

Homework 1

DATE: 14 Sep. 2026

DUE : 21 Sep. 2026

Textbook: Classical Mechanics, 3rd Ed. (H. Goldstein, C. Poole, and J. Safko)



PREAMBLE

Let me emphasize again the role of studying Classical Mechanics and its deep connection to the whole of physics. The main goal is not merely to compute the motion of particles and rigid bodies, but to understand the structure of dynamical theory itself: how the full content of Newton's laws can be recast through a single scalar function and a variational principle, and why that reformulation is what makes mechanics portable to every other domain. The subject is deceptively familiar, its equations were fixed long ago, yet the framework it bequeathed is the one in which we still write down every modern theory.

Classical mechanics is rich yet subtle, building a hierarchy of formulations that each expose more of the underlying architecture. Newtonian mechanics organizes motion through forces; Lagrangian mechanics replaces them with generalized coordinates, the principle of least action, and a direct link between symmetries and conservation laws; Hamiltonian mechanics recasts the dynamics in phase space, where canonical transformations, Poisson brackets, and Hamilton–Jacobi theory reveal its deep geometric structure. No single formulation is best for all problems, the power lies in moving fluently between them.

Classical mechanics is thus both a self-contained account of macroscopic motion and the scaffolding on which later physics is built: the action principle and Hamiltonian formalism pass almost intact into quantum mechanics, statistical mechanics, and field theory. Its regime of validity has limits, quantum effects at small scales, relativistic effects at high speeds; yet within them it is exact, and beyond them it remains the language in which the successor theories are written.

Note: Your Homework is accessible at the University Blackboard platform. The homework problems are mostly taken from your textbook. Completed homework will be accepted on the mentioned due date. Any submission after the due date will reduce your grade by 25 %. Your HWs should be submitted via email.

Problem 1:

A rocket of mass $m_1 + m_2$ is launched with a velocity whose horizontal and vertical components are u_x and u_y . At the highest point in its path the rocket explodes into two parts of mass m_1 and m_2 that separate in a horizontal direction in the original plane of motion. Show that the fragments strike the ground at a distance apart given by

$$D = \frac{u_y}{g} \sqrt{\frac{2(m_1 + m_2)KE}{m_1 m_2}},$$

where KE is the kinetic energy produced by the explosion. (Neglect air resistance, the mass of the explosive, any spinning motion of the fragments, and assume g is constant.)

Problem 2: (exercise 5 in Goldstein's textbook)

Two wheels of radius a are mounted on the ends of a common axle of length b such that the wheels rotate independently. The whole combination rolls without slipping on a plane. Show that there are two nonholonomic equations of constraint,

$$\cos \theta \, dx + \sin \theta \, dy = 0,$$

$$\sin \theta \, dx - \cos \theta \, dy = \frac{1}{2}a(d\phi + d\phi'),$$

(where θ , ϕ , and ϕ' have meanings similar to those in the problem of a single vertical disk, and (x, y) are the coordinates of a point on the axle midway between the two wheels) and one holonomic equation of constraint,

$$\theta = C - \frac{a}{b}(\phi - \phi'),$$

where C is a constant.

Problem 3: (exercise 18 in Goldstein's textbook)

A Lagrangian for a particular physical system can be written as

$$L' = \frac{m}{2} (a\dot{x}^2 + 2b\dot{x}\dot{y} + c\dot{y}^2) - \frac{K}{2} (ax^2 + 2bxy + cy^2),$$

where a , b , and c are arbitrary constants but subject to the condition that $b^2 - ac \neq 0$. What are the equations of motion? Examine particularly the two cases $a = 0 = c$ and $b = 0$, $c = -a$. What is the physical system described by the above Lagrangian? Show that the usual Lagrangian for this system as defined by Eq. (1.56) is related to L' by a point transformation (cf. Derivation 10). What is the significance of the condition on the value of $b^2 - ac$?

"Classical mechanics gave us a deterministic view of the world. Quantum mechanics, conversely, gives us a probabilistic view instead. According to Newton, if you know the cause of an event, you can predict the outcome. According to M. Born, you can only predict how likely that outcome will be."

Leonid V. Azaroff